



SQL numeric functions

An introduction to SQL functions

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What are SQL functions?

SQL functions are **built-in operations** that can be used to perform various **calculations, manipulations, or transformations** on data within an SQL **database**. Advantages of using these functions include:

01. Efficiency

They take advantage of **internal algorithms** and **data structures**, which results in faster query execution times and improved overall performance.

02. Reliability and compatibility

They adhere to the **SQL standards** and are **implemented consistently** across different database platforms.

03. Documentation and built-in support

They come with **extensive documentation** provided by the database management system vendors, making them **easier for developers to understand and use** effectively.

04. Extensive functionality

Database management systems provide a **wide range of built-in functions** that allow for complex calculations, string manipulations, and date and time operations.

05. Portability

SQL queries use **standard functions; therefore**, they can be **easily migrated or executed** on different database systems without significant modifications.

06. Ease of use

They are **readily available for use** without any additional configuration. This saves you from reinventing the wheel by implementing common operations from scratch.

Syntax of SQL functions

The **output of a function** in SQL is treated as a **derived column** and appears in the SELECT clause of a query. The general syntax of an SQL function is:

```
SELECT  
  Column1,  
  Function_name (Arguments) AS Named_result  
FROM  
  Database_name.Table_name
```

The diagram illustrates the syntax of an SQL function within a query. It shows a SELECT statement with two columns: 'Column1' and 'Function_name (Arguments) AS Named_result'. A red box highlights 'Function_name', with a red line pointing to a red callout box. A blue box highlights '(Arguments)', with a blue line pointing to a blue callout box. A purple box highlights 'AS Named_result', with a purple line pointing to a purple callout box. The 'FROM' clause is shown below the SELECT clause, with 'Database_name.Table_name' as the source.

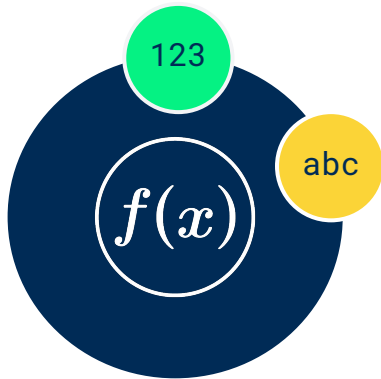
This defines the **specific SQL function** we want to use, such as SUM or COUNT .

These are the **input arguments** that the function requires. Some functions do not require any arguments while others require one or more arguments.

It is good practice to **name function results** descriptively with prefixes that explain the function, e.g. total_sales for SUM(sales).

Function types

Functions can be categorized by **what input** they take.



Numeric

These functions perform **calculations on a set of values** in a column and **return a numerical value**.

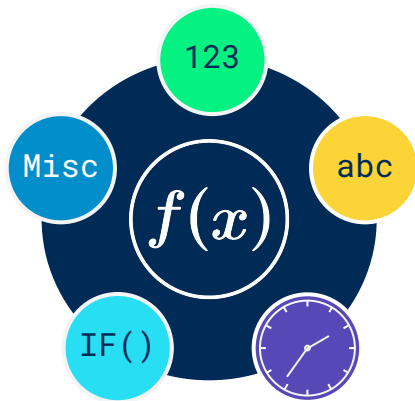
`SUM()`, `AVG()`, `COUNT()`, `MAX()`, `MIN()`, `POWER()`,
`SQRT()`, `ROUND()`

String

These functions **operate on string values** and perform operations such as concatenation, manipulation, and text formatting.

`CONCAT()`, `LENGTH()`, `SUBSTRING()`, `UPPER()`, `LOWER()`

Function types



Many functions do not fit this classification! For example, `MIN()` works with strings, numbers, and dates. Focus on **understanding** how **each function** works, rather than classifying them.

Datetime

These are functions used to **handle date and time values** and perform operations like formatting, extraction, and manipulation.

`CURRENT_DATE()`, `DATEADD()`, `FORMAT()`

Conditional flow

These functions allow for conditional logic in your SQL queries.

`CASE()`, `IF()`, `IIF()`

Miscellaneous

This category represents a variety of functions that **do a variety of things** such as converting data types and dealing with NULL values.

`CONVERT()`, `CAST()`, `NULLIF()`, `IFNULL()`

How functions behave

SQL functions vary in behaviour. **Aggregate functions** summarise data at a **column level**. **Scalar functions** manipulate data at the **row level**.

Patient_id	Born	FL00R(Born)
1	1969	1960
3	1972	1970
7	1954	1950
8	2004	2000

AVG(Born)
1973.5

Scalar functions take a row and return a single **value for each row**, e.g. UPPER(), LOWER(), CONVERT(), CAST(), IF(), FORMAT() and arithmetic operators (+, -, *, /)

Example:

FL00R(Born) rounds the values in each row down to the nearest integer. **Each row** has a **new** decade **column** with the output of FL00R().

Aggregate functions take a **set of rows** as input and return a **single summary value**, e.g. SUM(), AVG(), COUNT(), MAX(), MIN()

Example:

AVG(Born) calculates the average of the values in born, and returns a single summary value of the column.



Window functions are a third way in which functions can behave which is a “hybrid” of **aggregate** and **scalar** functions.

Example data

To find the **total amount of income expenditure** in the **Free State province** in South Africa, we are going to use the South African Household Income and Expenditure Survey dataset (SAHIES).

The table is named **Income_expenditure_2020** in the **Sahies** database.

Expenditure_group	Western_cape	Northern_cape	Free_state	KwaZulu-Natal	Northwest	...	Mpumalanga	Limpopo
Housing	16400	20000	24799	21200	21200	...	22799	23599
Recreation	1521	989	1255	1217	1179	...	1065	912
Transport	17974	15406	11983	13481	12625	...	12839	13481

Example: Aggregate functions

SUM() is an **aggregate function** that returns the total sum of a numeric column.

Query

```
SELECT
    SUM(Free_state) AS Total_free_state
FROM
    Sahies.Income_expenditure_2020;
```

Output

Total_free_state
38037

$$\text{SUM} \left[\begin{array}{|c|} \hline \text{Free_state} \\ \hline 24799 \\ \hline 1255 \\ \hline 11983 \\ \hline \end{array} \right] = 38037$$

We see that our query returns a **single value**, which is the sum of all rows of the **Free_state** column.

Example: Multiple aggregate functions

We can aggregate **multiple columns** or the same **column** using a **different function**.

Suppose we want to **calculate the total income spent** for the **Western** and **Free State** provinces and the **average spent** for the **Free State** province.

Query

```
SELECT
    SUM(Western_cape) AS Total_spent_western_cape,
    SUM(Free_state) AS Total_spent_Free_state,
    AVG(Free_state)
FROM
    Sahies.Income_expenditure_2020;
```

By not naming the average spent in the Free State, it is assigned a default name that may not be as descriptive.

Output

Total_spent_western_cape	Total_spent_Free_state	AVG(Free_state)
35895	38037	12679.0000

Example: Multiple aggregate functions

Since aggregate functions produce a single row, we **cannot use** them with the columns in our main table.

Query

```
SELECT
  Free_state,
  AVG(Free_state)
FROM
  Sahies.Income_expenditure_2020;
```

Free_state has 3 rows, and AVG(Free_state) has 1, so running this query results in an error.

Output



Error!

Example: Scalar functions

Suppose we want to calculate the **difference in income expenditure between the Free State** and the **Northern Cape** province.

Query

```
SELECT
    Expenditure_group,
    (Free_state
    - Northern_cape) AS Diff_fs_and_nc
FROM
    Sahies.Income_expenditure_2020;
```

Free_state	-	Northern_cape
24799	→	20000
1255	→	989
11983	→	15406

Output

Expenditure_group	Diff_fs_and_nc
Housing	4799
Recreation	266
Transport	-3423

Note how this name is slightly harder to understand because we used abbreviations.

We see that the **calculation occurred on a row level** because the results set includes a value for each row.

Using functions together

We can use **functions within functions**, known as **nesting**. SQL evaluates the **innermost** function **first** and then works its way **outwards**. The result of each inner function is used as the input for the outer function.

Query

```
SELECT
    SUM(Western_cape) AS Total_spent_western_cape,
    SUM(Northern_cape) AS Total_spent_northern_cape,
    ROUND(AVG(Free_state),0) AS Average_spent_free_state
FROM
    Sahies.Income_expenditure_2020;
```

ROUND() rounds a value to the specified amount of decimals.
AVG(Free_state) is calculated first, then rounded to 0 decimal places.

Output

Total_spent_western_cape	Total_spent_northern_cape	Average_spent_free_state
35895	36395	12679

SQL functions without input arguments

Some SQL built-in functions **do not require any arguments**. They are used to perform calculations or retrieve information.

CURRENT_DATE()

Returns the current date.

Query: **SELECT**
 CURRENT_DATE();

Output:

CURRENT_DATE()
2023-06-20

RAND()

Generates a random number between 0 and 1.

Query: **SELECT**
 Free_state,
 RAND() AS Random_number **FROM**...;

Output:

Limpopo	Random_number
23599	0.81247...
912	0.01824...
13481	0.55863...